Ho Ko

AI / ML SOFTWARE ENGINEER

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SUMMARY

AI/ML Software Engineer with cross-domain experience spanning astronomy, weather forecasting, economics, healthcare, NLP, and computer vision. Works across the full **model lifecycle** — training and distilling deep models, architecting highly efficient inference engines, and building scalable, high-throughput serving — and brings that intelligence **on-device** through WebGPU, Three.js, and local LLMs in privacy-preserving systems that have reached **1B+ users**.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Google — 2026–Present

- Build **knowledge-grounded retrieval & ranking models** on Google Lens that power AI Mode in Google Search and AI Overviews at Search scale.
- **Distill Gemini** into highly compressed models that hold near-equal quality while meeting the Lens stack's latency, scalability, and multimodal requirements.

Software Engineer, Google — 2023–2026

- Core contributor to the **LiteRT LM inference engine**, running LLMs privately and offline across Windows, macOS, Linux, and Android — including NPU acceleration.
- Enabled Gemma multimodality (vision + audio + text) and Gemini Nano on-device; helped ship **Chrome Built-in AI and Google AI Core, reaching 1B+ users**.
- Shipped the **Google AI Edge Gallery** app on Android and iOS.

Research Collaborator, Google DeepMind — 2024–2025

- Designed **Sparse RAG (ICLR 2025)**, a sparse decoding paradigm that encodes retrieved docs in parallel and attends only to the most relevant caches via learned control tokens.
- Cut long-range attention latency while **improving generation quality across 4 retrieval benchmarks**.

Research Collaborator, Google Health & Fitbit — 2024–2025

- Built and evaluated ML models **detecting hypertension risk from raw wrist PPG** passively collected on a consumer smartwatch (medRxiv), enabling non-invasive screening at population scale.

Software Engineer, MyCareLinq — 2023

- Built **NLP pipelines and a home-health-care knowledge graph** from unstructured health documentation.
- Integrated LLMs powering caregiving assistants that surface contextual recommendations and care-plan summaries.

Software Engineer Intern, Google — 2022

- Built a TFLite tensor extractor + metadata-writer with classification rules at **100% coverage of common GenAI task libraries** (LiteRT Task Library).
- Integrated a **C++ code generator** compiling optimized inference pipelines for arbitrary TFLite files.

Research Assistant, Caltech (SURF) — 2018

- Analyzed transient imaging from the Zwicky Transient Facility; **co-authored a Type IIb supernova study in ApJL (878:L5)** and simulated double-peak light-curve models.

EDUCATION

M.S. Computer Science · GPA 4.0/4.0
University of Southern California — AI & ML specialization
2021–2023

M.S. Electrical & Computer Eng. · GPA 3.9/4.0
University of Waterloo — Machine Learning & AI
2020–2021

B.S. Computer Science · GPA 3.8/4.0
National Central University, Taiwan
2016–2020

SELECTED PUBLICATIONS

- **Accelerating Inference of Retrieval-Augmented Generation via Sparse Context Selection** — **ICLR 2025** (Google DeepMind).
- **Opportunistic Hypertension Detection from Wearable PPG** — **medRxiv 2025** (Google Health & Fitbit).
- **Forecasting of Solar Power Generation using Real-time Meteorological Information** — **2020**.
- **A Type IIb Supernova (ZTF18aalrxas)** — **ApJL 878:L5** (2019).
- **Real-world WebAssembly Performance for ML** — **ACM IMC 2023**.

TECHNICAL SKILLS

Languages: Python, C/C++, Swift, Java, Kotlin, JavaScript, x86 Assembly, MATLAB
ML Lifecycle: Model Training, Distillation, Quantization & Compression, Inference Engines, On-Device Deployment, Scalable Serving
On-Device & Web: WebGPU, Three.js, Transformers.js, MediaPipe, WebRTC, Local LLMs (Gemma, LFM)
Domains: NLP, Computer Vision, Healthcare, Astronomy, Weather Forecasting, Economics